Particles and Nuclei 2

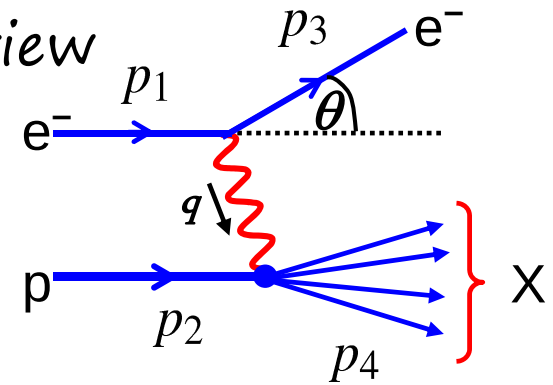
- 4/14/2026
- Announcements:

- HW 5 due Thurs class time
- Office Hours tomorrow 11:10am

- New Reading -- Rest of Griff Ch 9 (excluding 9.4,9.5, [9.1]) Griff Sections 10.7-10.9 (you don't need the rest to understand) incl Higg's Mechanism explanation of Ch 9 weak mixing angle, and Ch 11. (very short on neutrino oscillations)

DIS→Struct. Fn→Pdf(s)

- Review



X

$$\frac{d^2\sigma}{dxdQ^2} = \frac{4\pi\alpha^2}{Q^4}\left[(1-y)\frac{F_2(x,Q^2)}{x} + y^2 F_1(x,Q^2)\right]$$

$$\frac{d^2\sigma}{dE_3 d\Omega} = \frac{\alpha^2}{4E_1^2 \sin^4\theta/2}\left[\frac{1}{\nu}F_2(x,Q^2)\cos^2\frac{\theta}{2} + \frac{2}{M}F_1(x,Q^2)\sin^2\frac{\theta}{2}\right]$$

Still just t from propagator

- Callan Gross: $F_2(x) = 2xF_1(x)$ spin 1/2

q pdfs: $F_2^p(x) = x\sum_q e_q^2 q^p(x)$

Griff start: most general tensor form

$$K^{\mu\nu}_{\text{proton}} = -K_1 g^{\mu\nu} + \frac{K_2}{(Mc)^2}p^\mu p^\nu + \frac{K_4}{(Mc)^2}q^\mu q^\nu + \frac{K_5}{(Mc)^2}(p^\mu q^\nu + p^\nu q^\mu) \qquad (8.15)$$

$\nu_\mu p \to \mu^- X$

$$\frac{d^2\sigma^{\nu p}}{dxdy} = \frac{G_F^2 s}{2\pi}\left[(1-y)F_2^{\nu p}(x,Q^2) + y^2 xF_1^{\nu p}(x,Q^2) + y\left(1-\frac{y}{2}\right)xF_3^{\nu p}(x,Q^2)\right]$$

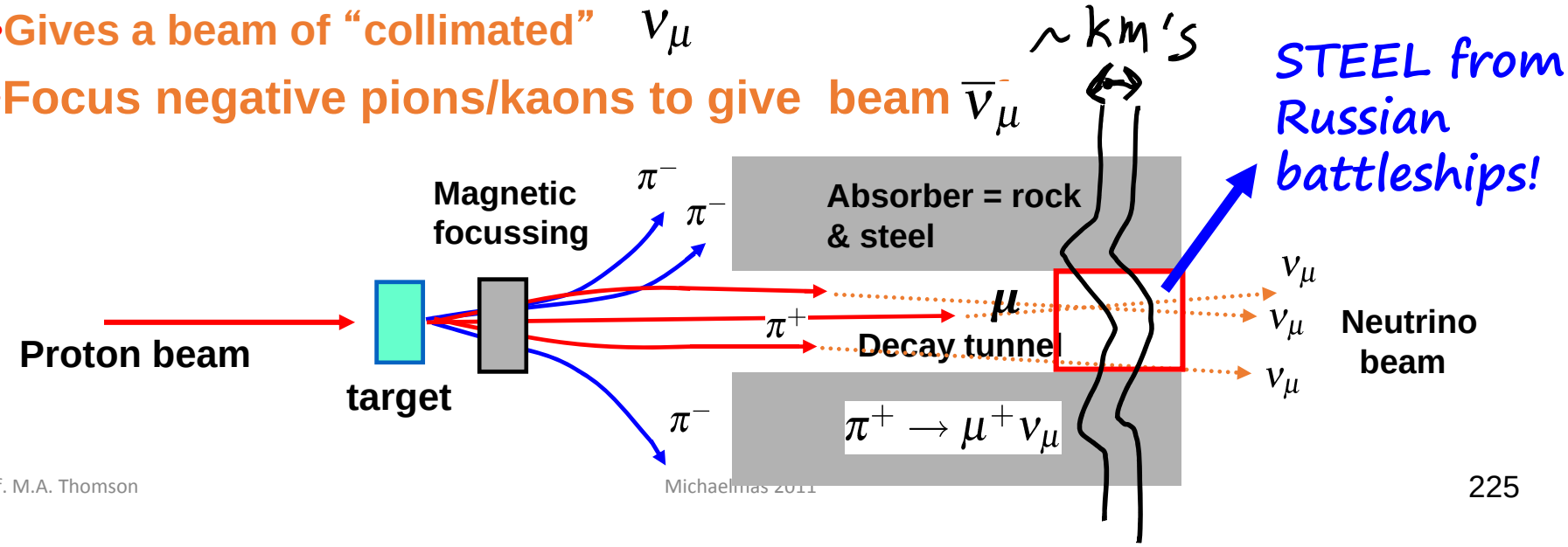
neutrino DIS → Important for sea quark measurements

Neutrino Scattering

- In handout 6 considered **electron-proton** Deep Inelastic Scattering where a virtual photon is used to probe nucleon structure
- Can also consider the weak interaction equivalent: **Neutrino Deep Inelastic Scattering** where a virtual W-boson probes the structure of nucleons
  - ➡ additional information about parton structure functions
  - + provides a good example of calculations of weak interaction cross section

★ **Neutrino Beams:**

- Smash high energy protons into a fixed target ➡ hadrons
- Focus positive pions/kaons
- Allow them to decay $\pi^+ \to \mu^+ \nu_\mu$ + $K^+ \to \mu^+ \nu_\mu \ (BR \approx 64\%)$
- Gives a beam of "collimated" ν_μ
- Focus negative pions/kaons to give beam $\overline{\nu}_\mu$



Weak Interaction Feynman Rules

From previous reading→ Chapter 9.1

- Propagators

$$\frac{-i(g_{\mu\nu} - q_\mu q_\nu / M^2c^2)}{q^2 - M^2c^2} \quad \text{(propagator for } W \text{ and } Z\text{)}$$

$$\frac{ig_{\mu\nu}}{(Mc)^2} \quad \text{(propagator for } q^2 \ll (Mc)^2\text{)}$$

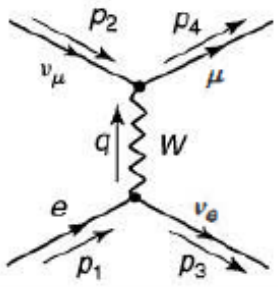
Interesting
No q^2 in denominator
(no s or t or u)

$$\frac{-ig_W}{2\sqrt{2}}\gamma^\mu(1 - \gamma^5) \quad \text{(weak vertex factor)}$$

Example 9.1 *Inverse Muon Decay* Consider the process

$$\nu_\mu + e^- \rightarrow \mu^- + \nu_e$$

represented (in lowest order) by the diagram



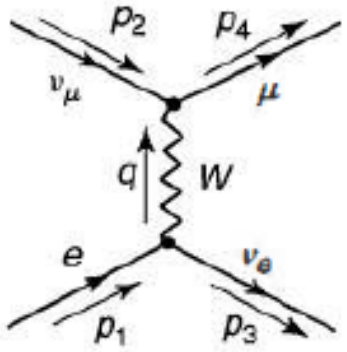
Example calc

- Similar to $\nu + q$ main way to detect neutrinos

Example 9.1 Inverse Muon Decay Consider the process

$$\nu_\mu + e^- \rightarrow \mu^- + \nu_e$$

represented (in lowest order) by the diagram



$$\sum_{\text{spins}} |\mathscr{M}|^2 = \left(\frac{g_w^2}{8(M_W c)^2}\right)^2 \mathrm{Tr}[\gamma^\mu(1-\gamma^5)(\not{p}_1 + m_e c)\gamma^\nu(1-\gamma^5)\not{p}_3]$$
$$\times \mathrm{Tr}[\gamma_\mu(1-\gamma^5)\not{p}_2\gamma_\nu(1-\gamma^5)(\not{p}_4 + m_\mu c)] \qquad (9.7)$$

- Not very different than QED calcs, but now the extra term with the $\gamma 5$ (but odd number of gammas 0, so…)

HW Last yr

9.2 Calculate the trace

$$\mathrm{Tr}\left[\gamma^{\mu}(c_V - c_A\gamma^5)(\not{p}_1 + m_1 c)\gamma^{\nu}(c_V - c_A\gamma^5)(\not{p}_2 + m_2 c)\right]$$

use pms

for arbitrary (real) numbers c_V and c_A.

$$\left[\text{Answer}: 4(c_V^2 + c_A^2)[p_1^{\mu}p_2^{\nu} + p_1^{\nu}p_2^{\mu} - p_1 \cdot p_2\, g^{\mu\nu}] + 4(c_V^2 - c_A^2)m_1 m_2 c^2 g^{\mu\nu} - 8ic_V c_A\, \epsilon^{\mu\nu\lambda\sigma} p_{1\lambda} p_{2\sigma}\right] \quad (9.159)$$

9.3 (a) Calculate $\langle |\mathcal{M}|^2 \rangle$ for $\nu_\mu + e^- \to \mu^- + \nu_e$ using the more general coupling $\gamma^\mu(1 + \epsilon\,\gamma^5)$. Check that your answer reduces to Equation 9.111 in the case $\epsilon = -1$.

$$\left[\text{Answer}: \frac{1}{4}\left(\frac{g_w}{\hbar c M_W c}\right)^4 [(1-\epsilon^2)^2 (p_1 \cdot p_4)(p_2 \cdot p_3) + (1 + 6\epsilon^2 + \epsilon^4)(p_1 \cdot p_2)(p_3 \cdot p_4)]\right]$$

$\to \frac{u^2}{4}$

$\to s^2/4$

(you may ignore masses as suggested by answer)
I think Griff explains that mass terms will not survive anyway, you do not need to use them in any case fro beginning , as we

HW Help Last yr

$$\mathscr{M} = \frac{g_w^2}{8(M_W c)^2}[\overline{u}(3)\gamma^\mu(1-\gamma^5)u(1)][\overline{u}(4)\gamma_\mu(1-\gamma^5)u(2)]$$

$$\sum_{\text{spins}} |\mathscr{M}|^2 = \left(\frac{g_w^2}{8(M_W c)^2}\right)^2 \mathrm{Tr}[\gamma^\mu(1-\gamma^5)(\not{p}_1 + m_e c)\gamma^\nu(1-\gamma^5)\not{p}_3]$$
$$\times \mathrm{Tr}[\gamma_\mu(1-\gamma^5)\not{p}_2\gamma_\nu(1-\gamma^5)(\not{p}_4 + m_\mu c)] \qquad (9.7)$$

$$= 4\left(c_V^2 + c_A^2\right)\left[p_1^\mu p_2^\nu + p_1^\nu p_2^\mu - (p_1 \cdot p_2)g^{\mu\nu}\right] + 4\left(c_V^2 - c_A^2\right)m_1 m_2 c^2 g^{\mu\nu}$$
$$- 8ic_V c_A \epsilon^{\mu\nu\lambda\sigma} p_{1\lambda} p_{2\sigma}.$$

Trace Theorems/Casimir's Trick

$$\mathcal{M} = -\frac{g_e^2}{(p_1 - p_3)^2}[\bar{u}(3)\gamma^\mu u(1)][\bar{u}(4)\gamma_\mu u(2)]$$

4 x2=8 summed matrices sandwiched with countless e.g. (32) norm factors

→ where 'Tr' denotes the *trace* of the matrix (the sum of its diagonal ele

$$\mathrm{Tr}(A) = \sum_i A_{ii}$$

$$\not{p}_a \equiv \gamma^\mu p_{\mu a}$$

Conclusion:

$$\sum_{\text{spins}} [\bar{u}(a)\Gamma_1 u(b)][\bar{u}(a)\Gamma_2 u(b)]^*$$

1. $\mathrm{Tr}(A+B) = \mathrm{Tr}(A) + \mathrm{Tr}(B)$
2. $\mathrm{Tr}(\alpha A) = \alpha\,\mathrm{Tr}(A)$
3. $\mathrm{Tr}(AB) = \mathrm{Tr}(BA)$

---

4. $g_{\mu\nu}g^{\mu\nu} = 4$
5. $\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu = 2g^{\mu\nu}$ 5′. $\not{a}\not{b} + \not{b}\not{a} = 2a\cdot b$
6. $\gamma_\mu\gamma^\mu = 4$
7. $\gamma_\mu\gamma^\nu\gamma^\mu = -2\gamma^\nu$ 7′. $\gamma_\mu\not{a}\gamma^\mu = -2\not{a}$

---

8. $\gamma_\mu\gamma^\nu\gamma^\lambda\gamma^\mu = 4g^{\nu\lambda}$ 8′. $\gamma_\mu\not{a}\not{b}\gamma^\mu = 4(a\cdot b)$
9. $\gamma_\mu\gamma^\nu\gamma^\lambda\gamma^\sigma\gamma^\mu = -2\gamma^\sigma\gamma^\lambda\gamma^\nu$ 9′. $\gamma_\mu\not{a}\not{b}\not{c}\gamma^\mu = -2\not{c}\not{b}\not{a}$

llection of 'trace theorems':

0. The trace of the product of an *odd* number of gamma matrices is *zero*.
1. $\mathrm{Tr}(1) = 4$
2. $\mathrm{Tr}(\gamma^\mu\gamma^\nu) = 4g^{\mu\nu}$ 12′. $\mathrm{Tr}(\not{a}\not{b}) = 4(a\cdot b)$
3. $\mathrm{Tr}(\gamma^\mu\gamma^\nu\gamma^\lambda\gamma^\sigma)$ 13′. $\mathrm{Tr}(\not{a}\not{b}\not{c}\not{d})$

$= 4(g^{\mu\nu}g^{\lambda\sigma} - g^{\mu\lambda}g^{\nu\sigma} + g^{\mu\sigma}g^{\nu\lambda})$ $= 4(a\cdot b\,c\cdot d - a\cdot c\,b\cdot d + a\cdot d\,b\cdot c)$

γ, since $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$ is the product of an *even* number of γ matrices, it rom Rule 10 that $\mathrm{Tr}(\gamma^5\gamma^\mu) = \mathrm{Tr}(\gamma^5\gamma^\mu\gamma^\nu\gamma^\lambda) = 0$. When γ^5 is multiplied y an *even* number of γ's, we find

14. $\mathrm{Tr}(\gamma^5) = 0$
15. $\mathrm{Tr}(\gamma^5\gamma^\mu\gamma^\nu) = 0$ 15′. $\mathrm{Tr}(\gamma^5\not{a}\not{b}) = 0$
16. $\mathrm{Tr}(\gamma^5\gamma^\mu\gamma^\nu\gamma^\lambda\gamma^\sigma) = 4i\epsilon^{\mu\nu\lambda\sigma}$ 16′. $\mathrm{Tr}(\gamma^5\not{a}\not{b}\not{c}\not{d}) = 4i\epsilon^{\mu\nu\lambda\sigma}a_\mu b_\nu c_\lambda d_\sigma$

note about γ^5, ϵ^{abcd}

8. $\gamma_\mu \gamma^\nu \gamma^\lambda \gamma^\mu = 4g^{\nu\lambda}$ 8′. $\gamma_\mu \not{a} \not{b} \gamma^\mu = 4(a \cdot b)$

9. $\gamma_\mu \gamma^\nu \gamma^\lambda \gamma^\sigma \gamma^\mu = -2\gamma^\sigma \gamma^\lambda \gamma^\nu$ 9′. $\gamma_\mu \not{a} \not{b} \not{c} \gamma^\mu = -2\not{c} \not{b} \not{a}$

nd a collection of 'trace theo

10. The trace of the p

11. $\mathrm{Tr}(1)$

12. $\mathrm{Tr}(\gamma^\mu \gamma^\nu)$

13. $\mathrm{Tr}(\gamma^\mu \gamma^\nu \gamma^\lambda \gamma^\sigma) = 4(g^{\mu\nu} g^{\lambda\sigma} - g^{\mu\lambda} g^{\nu\sigma} + g^{\mu\sigma} g^{\nu\lambda})$ 13′. $\mathrm{Tr}(\not{a} \not{b} \not{c} \not{d}) = 4(a \cdot b\, c \cdot d - a \cdot c\, b \cdot d + a \cdot d\, b \cdot c)$

$$\epsilon^{\mu\nu\lambda\sigma} = \begin{cases} -1, & \text{if } \mu\nu\lambda\sigma \text{ is an even permutation of } 0123 \\ +1, & \text{if } \mu\nu\lambda\sigma \text{ is an odd permutation.} \\ 0, & \text{if any two indices are the same} \end{cases}$$

$\epsilon^{\mu\nu\lambda\sigma}$

$\epsilon^{1023} = -1$

$\delta_{\mu\nu}$ or $\delta_{\mu\nu}$ times $\epsilon^{\cdots} = 0$

Finally, since $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$ is the product of an *even* number of γ matrices, it follows from Rule 10 that $\mathrm{Tr}(\gamma^5\gamma^\mu) = \mathrm{Tr}(\gamma^5\gamma^\mu\gamma^\nu\gamma^\lambda) = 0$. When γ^5 is multiplied by an *even* number of γ's, we find

14. $\mathrm{Tr}(\gamma^5) = 0$

15. $\mathrm{Tr}(\gamma^5\gamma^\mu\gamma^\nu) = 0$ 15′. $\mathrm{Tr}(\gamma^5 \not{a} \not{b}) = 0$

16. $\mathrm{Tr}(\gamma^5\gamma^\mu\gamma^\nu\gamma^\lambda\gamma^\sigma) = 4i\epsilon^{\mu\nu\lambda\sigma}$ 16′. $\mathrm{Tr}(\gamma^5 \not{a} \not{b} \not{c} \not{d}) = 4i\epsilon^{\mu\nu\lambda\sigma} a_\mu b_\nu c_\lambda d_\sigma$

HW Help Last year

- Start w/

$$\mathcal{M} \cong \left(\frac{g_w^2}{8(M_W c)^2}\right)[\bar{u}(3)\gamma^\mu(1-\gamma^5)u(1)][\bar{u}(4)\gamma_\mu(1-\gamma^5)u(2)]$$

$$\bar{u}_2 \not{p}_2 (1-\gamma^5) u_4 \quad \bar{u}_1 (1-\gamma^5) u_3$$

$$\bar{u}_1 \not{p}_1 (1-\gamma^5) \not{p}_3 \gamma^\nu (1-\gamma^5) u_1 \quad \bar{u}_4 \gamma_\mu (1-\gamma^5) \not{p}_2 \gamma_\nu (1-\gamma^5)$$

$$C_A^4 \left(\mathrm{Tr}[\gamma^\mu (1-\gamma^5) \not{p}_3 \gamma^\nu (1-\gamma^5) \not{p}_1] \times \mathrm{Tr}[\gamma_\mu (1-\gamma^5) \not{p}_2 \gamma_\nu (1-\gamma^5) \not{p}_4] \right)$$

$\varepsilon = C_V / C_A$ $\quad 1-\varepsilon\gamma^5 \quad 1-\varepsilon\gamma^5 \quad 1-\varepsilon\gamma^5 \quad 1-\varepsilon\gamma^5$

For these traces you can use identity from Problem 9.2 (Griffiths)
[you don't need to prove it]

- Identity from slide 2 written in two forms

$$
\begin{aligned}
&4c_V^2\Big\{(1+\epsilon^2)\left[p_1^\mu p_2^\nu + p_1^\nu p_2^\mu - (p_1\cdot p_2)g^{\mu\nu}\right] - 2i\epsilon(p_{1\lambda}p_{2\sigma})\epsilon^{\mu\nu\lambda\sigma} \\
&\quad + m_1 m_2 c^2 g^{\mu\nu}(1-\epsilon^2)\Big\} \\
&= 4\left(c_V^2 + c_A^2\right)\left[p_1^\mu p_2^\nu + p_1^\nu p_2^\mu - (p_1\cdot p_2)g^{\mu\nu}\right] + 4\left(c_V^2 - c_A^2\right) m_1 m_2 c^2 g^{\mu\nu} \\
&\quad - 8ic_V c_A \epsilon^{\mu\nu\lambda\sigma} p_{1\lambda} p_{2\sigma}.
\end{aligned}
$$

form given in prob 9.2

replace 1, 2 → 3, 1 ↓ → 2, 4

from prev ans slide

- Also the above is for traces with all γ^μ indices raised (contra)
- YOU MAY ASSUME (without proof) for all lowered gamma indicies (covariant) everything up reverses to down including for the Levi Civita 4tensor $\epsilon^{abdc} \longrightarrow \epsilon_{abcd}$ see later:

HW Help (Last yr)

$\bar{u}_1 \gamma^\mu (1-\gamma^5) \not{p}_3 \gamma^\nu (1-\gamma^5) u_1 \quad \bar{u}_4 \sigma_\mu (1-\gamma^5) \not{p}_2 \gamma_\nu (1-\gamma^5)$

$C_V^4 \left(\mathrm{Tr}[\gamma^\mu (1-\gamma^5) \not{p}_3 \gamma^\nu (1-\gamma^5) \not{p}_1] \times \mathrm{Tr}[\gamma_\mu (1-\gamma^5) \not{p}_2 \gamma_\nu (1-\gamma^5) \not{p}_4] \right)$

$\varepsilon = C_V / C_A$ $\quad 1-\varepsilon\gamma^5 \quad 1-\varepsilon\gamma^5 \quad 1-\varepsilon\gamma^5 \quad 1-\varepsilon\gamma^5$

$$= 4^2 \left(\frac{G_W}{8 M_W}\right)^2 (C_V^4) \{ [p_1^\mu p_3^\nu + p_3^\mu p_1^\nu - p_1 \cdot p_3 g^{\mu\nu}] - 2i\varepsilon p_{1\lambda} p_{3\sigma} \epsilon^{\mu\nu\lambda\sigma}] \times (1+\varepsilon^2)$$

$$\times [p_{2\mu} p_{4\nu} + p_{2\nu} p_{4\mu} - p_2 \cdot p_4 g_{\mu\nu}] - 2i\varepsilon p_{2\tau} p_{4\gamma} \epsilon_{\mu\nu\tau\gamma} \} \quad \pm 2 \times (1+\varepsilon^2)$$

These red indices should be chosen as different, because they otherwise can not be contracted with the first trace's since they are not related like the μ and ν are

There are 16 terms [without expanding tensor Einstein summations] after multiplication but better to group first two in each line together as on next slide

Remove terms

- terms as shown and remove some of the cross terms in the product of the two lines

$$= 4^2 \left(\frac{G_N}{8 M_W}\right)^2 (C_V 4)^2 \Big\{ \Big[\underbrace{p_1^\mu p_3^\nu + p_3^\mu p_1^\nu}_{1\ \text{group}} - p_1\cdot p_3\, g^{\mu\nu} \Big]^{*\,1+\varepsilon^2} - 2i\varepsilon\, p_{1\lambda} p_{3\sigma} \epsilon^{\mu\nu\lambda\sigma} \Big]$$

$$\times \Big[\underbrace{p_{2\mu} p_{4\nu} + p_{2\nu} p_{4\mu}}_{4} - p_2\cdot p_4\, g_{\mu\nu} \Big]^{*\,1+\varepsilon^2} - 2i\varepsilon\, p_{2\tau} p_{4\gamma} \epsilon_{\mu\nu\tau\gamma} \Big\}$$

(2) $\pm/2$; (5) $\pm/-2$; (6)

$g^{\mu\nu}\epsilon_{\mu\nu\chi\chi} = 0$ Also $2\times 5 \Rightarrow g^{\mu\nu} g_{\mu\nu} = 4 \Rightarrow = t^2$

~~2x6 and 5x3 are 0 because~~

1x6 and 3x4 are 0 because e.g. every term in Einstein sum expansion of 2nd term will appear w/ opposite sign

$$p_1^\mu p_3^\nu \epsilon^{\mu\nu\chi\chi} + p_1^\nu p_3^\mu \epsilon^{\mu\nu\chi\chi}$$

$$= p_1^\mu p_3^\nu \epsilon^{\mu\nu\chi\chi} - p_1^\mu p_3^\nu \epsilon^{\mu\nu\chi\chi} = 0$$

Remove terms

- terms as shown and remove some of the cross terms in the product of the two lines

$$= 4^2 \left(\frac{g^2}{8 M_W}\right)^2 (C_V^4) \Big\{ \Big[\underbrace{p_1^\mu p_3^\nu + p_3^\mu p_1^\nu}_{1\ \text{spin}\ \rho} - \underbrace{p_1 \cdot p_3}_{2}\, g^{\mu\nu} \Big]^{*1+\varepsilon^2}_{\pm/2} \underbrace{- 2i\varepsilon p_{1\lambda} p_{3\sigma} \epsilon^{\mu\nu\lambda\sigma}}_{3} \Big]$$

$$\times \Big[\underbrace{p_{2\mu} p_{4\nu} + p_{2\nu} p_{4\mu}}_{4} - \underbrace{p_2 \cdot p_4}_{5}\, g_{\mu\nu} \Big]^{*1+\varepsilon^2} \underbrace{- 2i\varepsilon p_{2\tau} p_{4\gamma} \epsilon_{\mu\nu\tau\gamma}}_{6}$$

also 1x5 & 2x 4 are $= -t^2/2$ b.c. $- p_1^\mu p_3^\nu \underbrace{\frac{t}{-2}}_{-2} g_{\mu\nu} - \underbrace{(p_1 \cdot p_3)(p_2 \cdot 4)}_{-t^2/4} + \text{same for } p_1^\nu p_3^\mu$

- So

$$= \underbrace{1x4}_{-t^2/2} + \cancel{1x5} + \cancel{1x6}^{0} + \underbrace{2x4}_{-t^2/2} + \cancel{2x5}^{+t^2} + \cancel{2x6}^{0} + \cancel{3x4}^{0} + \cancel{3x5}^{0} + 3x6$$

- t^2 terms cancel ? (check?)

Multiply out remaining terms

$$= 4^2 \left(\frac{g_W}{8 M_W}\right)^2 (C_V^4)^2 \Big\{ \Big[\underbrace{p_1^\mu p_3^\nu + p_3^\mu p_1^\nu}_{1\ \text{group}} - \underbrace{p_1 \cdot p_3}_{2}\, \underbrace{g^{\mu\nu}}_{3} \Big] - 2i\varepsilon\, p_{1\lambda} p_{3\sigma} \epsilon^{\mu\nu\lambda\sigma} \Big]$$

$$\times \Big[\underbrace{p_{2\mu} p_{4\nu} + p_{2\nu} p_{4\mu}}_{4} - \underbrace{p_2 \cdot p_4}_{5}\, g_{\mu\nu} - 2i\varepsilon\, p_{2\tau} p_{4\gamma} \epsilon_{\mu\nu\tau\gamma} \Big]$$

$$= \text{const} \Big\{ \underbrace{0}_{?} + \Big[\underbrace{p_1^\mu p_3^\nu p_{2\mu} p_{4\nu} + \ldots}_{1\times 4} \Big] * (1+\varepsilon^2)^2$$

$$(p_1 \cdot p_2)(p_3 \cdot 4) = s^2$$

$$+ 4 i^2 \varepsilon^2 \epsilon^{\mu\nu\lambda\sigma} \epsilon_{\mu\nu\tau\gamma} p_{1\lambda} p_{3\sigma} p_{2\tau} p_{4\gamma} \Big\}_{3\times 6}$$

± 2

$*(1+\varepsilon^2)$

- For the last term use (3x6) something from a problem in Chapter 7, Griffiths → Problem 7.35, see next slide

Identity in Griff for ϵ^{abcd}s

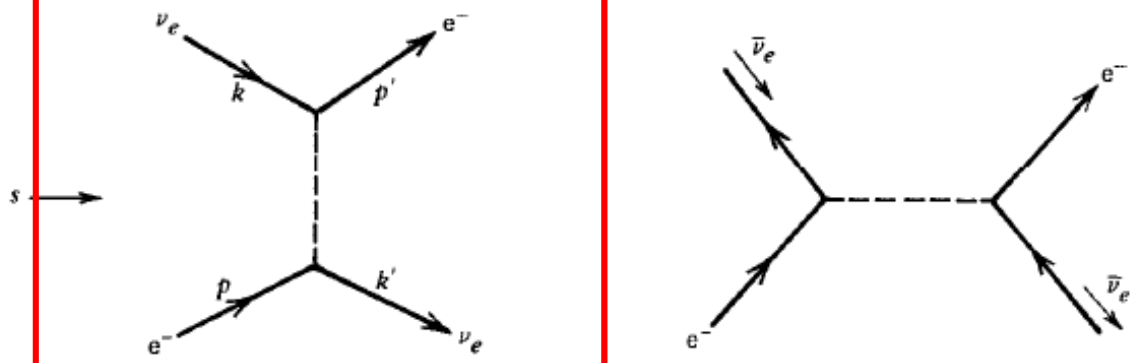
- Griff Ch 7 problem 7.35

7.35 (a) Show that $\epsilon^{\mu\nu\lambda\sigma}\epsilon_{\mu\nu\lambda\tau} = -6\,\delta^{\sigma}_{\tau}$ (summation over μ, ν, λ implied).
(b) Show that $\epsilon^{\mu\nu\lambda\sigma}\epsilon_{\mu\nu\theta\tau} = -2(\delta^{\lambda}_{\theta}\,\delta^{\sigma}_{\tau} - \delta^{\lambda}_{\tau}\,\delta^{\sigma}_{\theta})$.
(c) Find the analogous formula for $\epsilon^{\mu\nu\lambda\sigma}\epsilon_{\mu\phi\theta\tau}$.
(d) Find the analogous formula for $\epsilon^{\mu\nu\lambda\sigma}\epsilon_{\omega\phi\theta\tau}$.
[Here δ^{μ}_{ν} is the Kronecker delta: 1 if $\mu = \nu$, 0 otherwise. It can also be written in terms of the mixed (co/contravariant) metric tensor: $\delta^{\mu}_{\nu} = g^{\mu}{}_{\nu} = g_{\nu}{}^{\mu}$.]

- You can take it from here...

Halzen Another similar calc

- ve→ve (< ~80 GeV parton/electron beam energies)



(a) Neutrino-electron scattering (b) Antineutrino-electron scattering

Fig. 12.8 Charged current contributions to elastic $\nu_e e^-$ and $\bar{\nu}_e e^-$ scattering.

initial e$^-$ gives

$$\frac{1}{2}\sum_{\text{spins}}|\mathcal{M}|^2 = \frac{G^2}{4}\,\mathrm{Tr}\left(\gamma^\mu(1-\gamma^5)\not{p}\gamma^\nu(1-\gamma^5)\not{k}'\right)\mathrm{Tr}\left(\gamma_\mu(1-\gamma^5)\not{k}\gamma_\nu(1-\gamma^5)\not{p}'\right)$$

$$= 64G^2(k\cdot p)(k'\cdot p')$$

$$= 16G^2s^2 \qquad (12.57)$$

$$\frac{d\sigma(\nu_e e^-)}{d\Omega} = \frac{1}{64\pi^2 s}\overline{|\mathcal{M}|^2} = \frac{G^2 s}{4\pi^2}.$$

Quite Simple form

INTERESTING

t channel diagram

no longer keep propagator term
$q^2 \rightarrow t$

b.c. of W mass!

only s term in "numerator" is left

No angular dep b.c. spin is S= 0

Cool other diagram "cross" s→ t

- Use famous switch / crossing (now s diagram gets t)

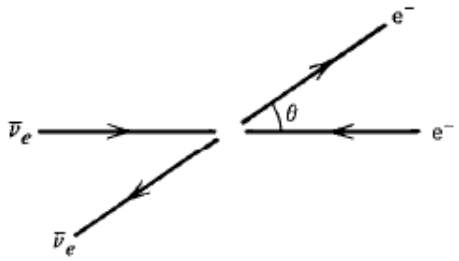


Fig. 12.9 Definition of θ for $\bar{\nu}_e e^-$ scattering.

The Feynman diagram for $\bar{\nu}_e e^- \to e^- \bar{\nu}_e$ is shown in Fig. 12.8b. We see that it can be obtained by crossing the neutrinos in $\nu_e e^- \to e^- \nu_e$ of diagram (a); see Sections 4.6 and 4.7. We therefore simply replace s by t in (12.57):

$$\frac{1}{2}\sum_{\text{spins}} |\mathcal{M}|^2 = 16G^2t^2$$

$$= 4G^2s^2(1 - \cos\theta)^2, \qquad (12.61)$$

where θ is the angle between the incoming $\bar{\nu}_e$ and the outgoing e^- (see Fig. 12.9), and

$$t \simeq -\frac{s}{2}(1 - \cos\theta),$$

see (4.45). From (12.61), we obtain

$$\frac{d\sigma(\bar{\nu}_e e^-)}{d\Omega} = \frac{G^2 s}{16\pi^2}(1 - \cos\theta)^2, \qquad (12.62)$$

Exercise

- Reminder : gamma matrices $\bar{\psi}\gamma^5\psi$ with $\gamma^5 \equiv i\gamma^0\gamma^1\gamma^2\gamma^3$

$$\gamma^5 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$ in the usual 2x2 form

- Alt convention (Weyl) for gamma

= (-1) x (our γ^0)

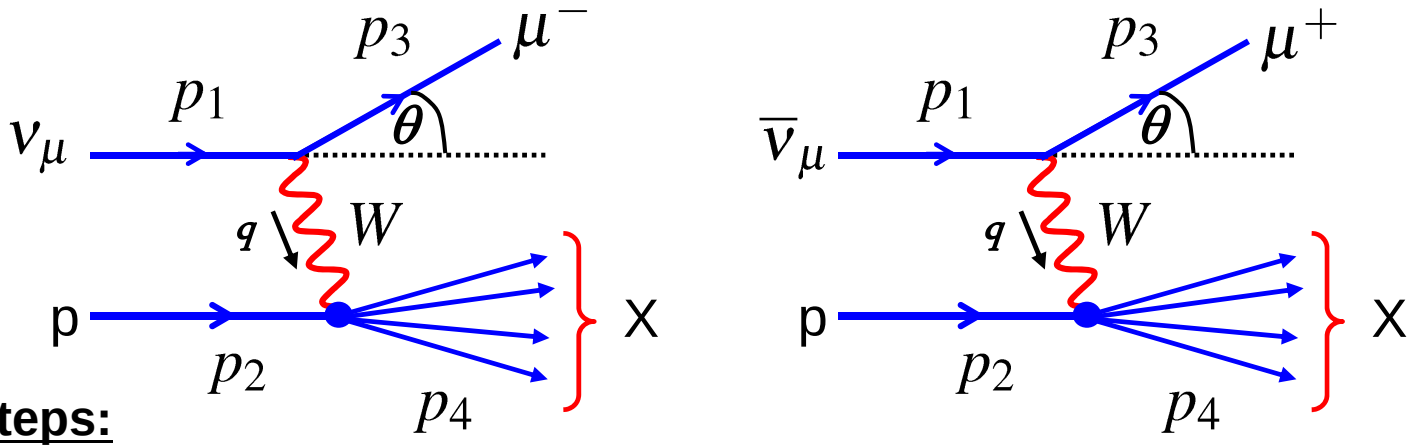
$$\gamma^0 = \begin{pmatrix} 0 & 1_2 \\ 1_2 & 0 \end{pmatrix}, \quad \gamma^i = \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix}, \quad \Rightarrow \quad \gamma^5 = \begin{pmatrix} -1_2 & \\ & 1_2 \end{pmatrix}.$$

- Other Related Operators

$$\vec{S} = \frac{1}{2}\vec{\Sigma} = \frac{1}{2}\begin{pmatrix} \vec{\sigma} & 0 \\ 0 & \vec{\sigma} \end{pmatrix}$$

- In Weyl convention: give form of $1+\gamma^5$ and $1-\gamma^5$
- In our convention show evaluate $(1+\gamma^5)(1-\gamma^5)$ (using algebra)

Appendix II: Deep-Inelastic Neutrino Scattering



Two steps:

- **First write down most general cross section in terms of structure functions**
- **Then evaluate expressions in the quark-parton model**

QED Revisited

★**In the limit $s \gg M^2$ the most general electro-magnetic deep-inelastic cross section (from single photon exchange) can be written (Eq. 2 of handout 6)**

$$\frac{\mathrm{d}^2\sigma_{e^\pm p}}{\mathrm{d}x\mathrm{d}Q^2} = \frac{4\pi\alpha^2}{Q^4}\left[(1-y)\frac{F_2(x,Q^2)}{x} + y^2 F_1(x,Q^2)\right]$$

- **For neutrino scattering typically measure the energy of the produced muon $E_\mu = E_\nu(1-y)$ and differential cross-sections expressed in terms of $\mathrm{d}x\mathrm{d}y$**
- **Using** $Q^2 = (s-M^2)xy \approx sxy$ ⇒ $$\frac{\mathrm{d}^2\sigma}{\mathrm{d}x\mathrm{d}y} = \left|\frac{\mathrm{d}Q^2}{\mathrm{d}y}\right| \frac{\mathrm{d}^2\sigma}{\mathrm{d}x\mathrm{d}Q^2} = sx\frac{\mathrm{d}^2\sigma}{\mathrm{d}x\mathrm{d}Q^2}$$

♦ In the limit $s \gg M^2$ the general Electro-magnetic DIS cross section can be written

$$\frac{d^2\sigma^{e^\pm p}}{dxdy} = \frac{4\pi\alpha^2 s}{Q^4}\left[(1-y)F_2(x,Q^2) + y^2 x F_1(x,Q^2)\right]$$

•**NOTE:** This is the most general **Lorentz Invariant** parity conserving expression

★For neutrino DIS parity is violated and the general expression includes an additional term to allow for parity violation. New structure function $F_3(x,Q^2)$

$$\nu_\mu p \to \mu^- X \qquad \frac{d^2\sigma^{\nu p}}{dxdy} = \frac{G_F^2 s}{2\pi}\left[(1-y)F_2^{\nu p}(x,Q^2) + y^2 x F_1^{\nu p}(x,Q^2) + y\left(1-\frac{y}{2}\right)xF_3^{\nu p}(x,Q^2)\right]$$

•For anti-neutrino scattering new structure function enters with opposite sign

$$\bar{\nu}_\mu p \to \mu^+ X \qquad \frac{d^2\sigma^{\bar{\nu} p}}{dxdy} = \frac{G_F^2 s}{2\pi}\left[(1-y)F_2^{\bar{\nu} p}(x,Q^2) + y^2 x F_1^{\bar{\nu} p}(x,Q^2) - y\left(1-\frac{y}{2}\right)xF_3^{\bar{\nu} p}(x,Q^2)\right]$$

•Similarly for neutrino-neutron scattering

$$\nu_\mu n \to \mu^- X \qquad \frac{d^2\sigma^{\nu n}}{dxdy} = \frac{G_F^2 s}{2\pi}\left[(1-y)F_2^{\nu n}(x,Q^2) + y^2 x F_1^{\nu n}(x,Q^2) + y\left(1-\frac{y}{2}\right)xF_3^{\nu n}(x,Q^2)\right]$$

$$\bar{\nu}_\mu n \to \mu^+ X \qquad \frac{d^2\sigma^{\bar{\nu} n}}{dxdy} = \frac{G_F^2 s}{2\pi}\left[(1-y)F_2^{\bar{\nu} n}(x,Q^2) + y^2 x F_1^{\bar{\nu} n}(x,Q^2) - y\left(1-\frac{y}{2}\right)xF_3^{\bar{\nu} n}(x,Q^2)\right]$$

Neutrino Interaction Structure Functions

★**In terms of the parton distribution functions we found (2) :**

$$\frac{d^2\sigma^{\nu p}}{dxdy} = \frac{G_F^2}{\pi} sx\left[d(x) + (1-y)^2\bar{u}(x)\right]$$

•**Compare coefficients of y with the general Lorentz Invariant form (p.321) and assume Bjorken scaling, i.e.** $F(x,Q^2) \to F(x)$

$$\frac{d^2\sigma^{\nu p}}{dxdy} = \frac{G_F^2 s}{2\pi}\left[(1-y)F_2^{\nu p}(x) + y^2 x F_1^{\nu p}(x) + y\left(1-\frac{y}{2}\right)xF_3^{\nu p}(x)\right]$$

•**Re-writing (2)** $\frac{d^2\sigma^{\nu p}}{dxdy} = \frac{G_F^2}{2\pi} s\left[2xd(x) + 2x\bar{u}(x) - 4xy\bar{u}(x) + 2xy^2\bar{u}(x)\right]$

and equating powers of y

$$\left.\begin{aligned} 2xd + 2x\bar{u} &= F_2 \\ -4x\bar{u} &= -F_2 + xF_3 \\ 2\bar{u} &= F_1 - xF_3/2 \end{aligned}\right\}$$

gives:

$$\boxed{\begin{aligned} F_2^{\nu p} = 2xF_1^{\nu p} &= 2x[d(x) + \bar{u}(x)] \\ xF_3^{\nu p} &= 2x[d(x) - \bar{u}(x)] \end{aligned}}$$

NOTE: again we get the **Callan-Gross** relation $F_2 = 2xF_1$

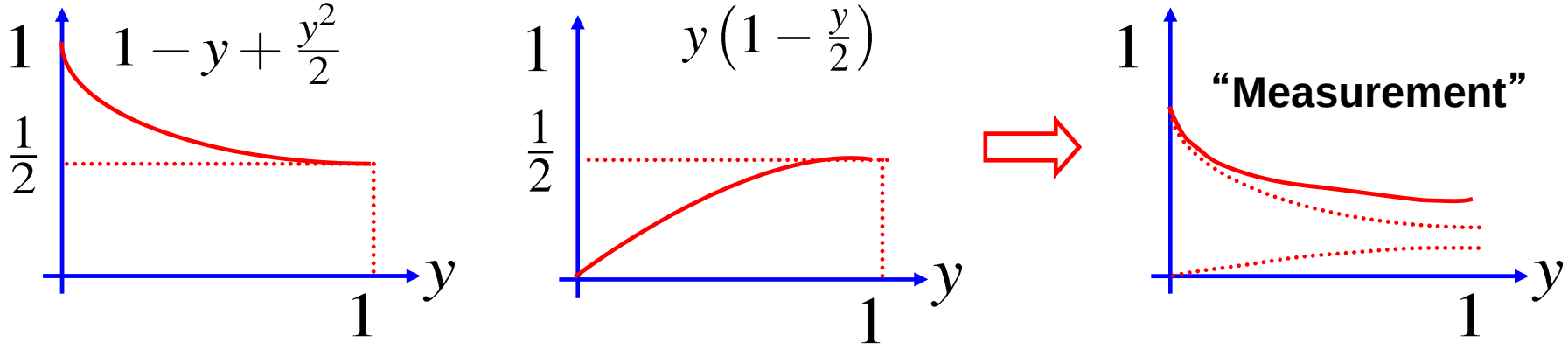
No surprise, underlying process is scattering from point-like spin-1/2 quarks

★ **Substituting back in to expression for differential cross section:**

$$\frac{d^2\sigma^{\nu p}}{dxdy} = \frac{G_F^2 s}{2\pi}\left[\left(1-y+\frac{y^2}{2}\right)F_2^{\nu p}(x) + y\left(1-\frac{y}{2}\right)xF_3^{\nu p}(x)\right]$$

★ **Experimentally measure** F_2 **and** F_3 **from** y **distributions at fixed** x

- **Different y dependencies (from different rest frame angular distributions) allow contributions from the two structure functions to be measured**



★ **Then use** $F_2^{\nu p} = 2x[d(x)+\bar{u}(x)]$ **and** $F_3^{\nu p} = 2[d(x)-\bar{u}(x)]$

➡ **Determine** $d(x)$ **and** $\bar{u}(x)$ **separately**

★**Neutrino experiments require large detectors (often iron) i.e. isoscalar target**

$$F_2^{\nu N} = 2xF_1^{\nu N} = \tfrac{1}{2}\left(F_2^{\nu p} + F_2^{\nu n}\right) = x[u(x) + d(x) + \overline{u}(x) + \overline{d}(x)]$$

$$xF_3^{\nu N} = \tfrac{1}{2}\left(xF_3^{\nu p} + xF_3^{\nu n}\right) = x[u(x) + d(x) - \overline{u}(x) - \overline{d}(x)]$$

★**For electron – nucleon scattering:**

$$F_2^{ep} = 2xF_1^{ep} = x[\tfrac{4}{9}u(x) + \tfrac{1}{9}d(x) + \tfrac{4}{9}\overline{u}(x) + \tfrac{1}{9}\overline{d}(x)]$$

$$F_2^{en} = 2xF_1^{en} = x[\tfrac{4}{9}d(x) + \tfrac{1}{9}u(x) + \tfrac{4}{9}\overline{d}(x) + \tfrac{1}{9}\overline{u}(x)]$$

•**For an isoscalar target**

$$F_2^{eN} = \tfrac{1}{2}\left(F_2^{ep} + F_2^{en}\right) = \tfrac{5}{18}x[u(x) + d(x) + \overline{u}(x) + \overline{d}(x)]$$

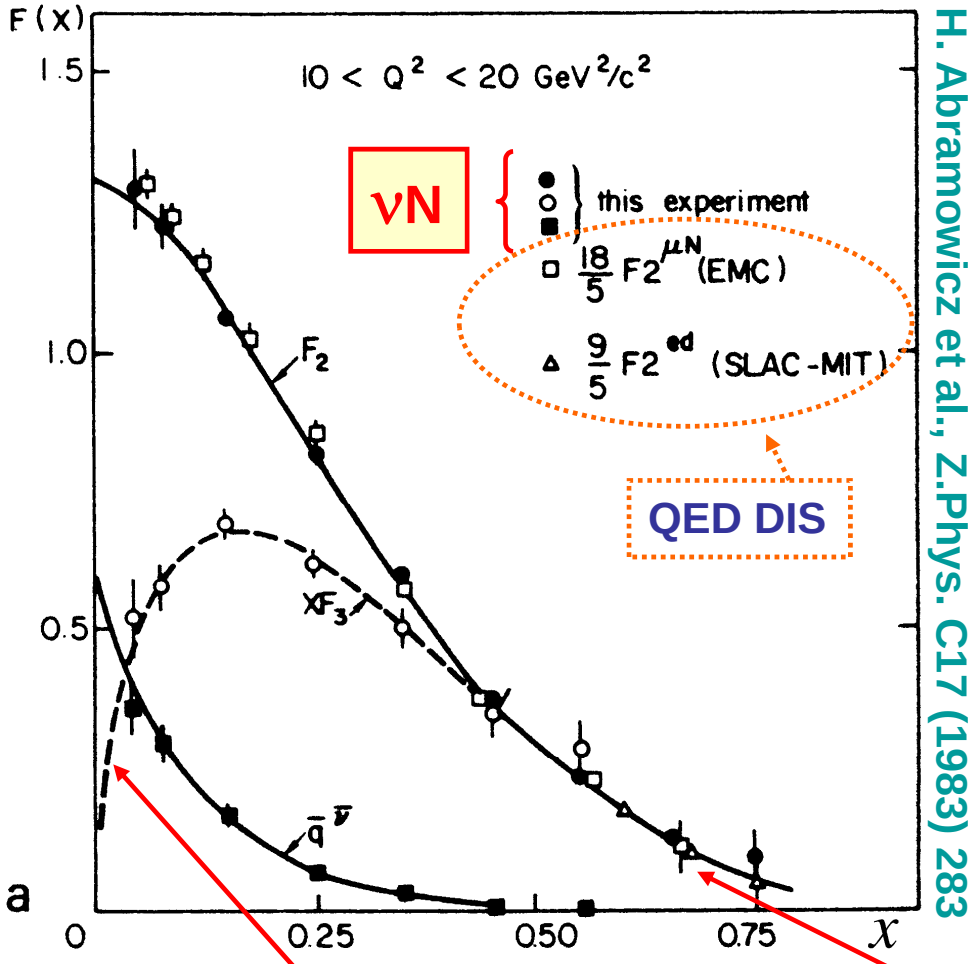
$$\Rightarrow \quad F_2^{\nu N} = \tfrac{18}{5}F_2^{eN}$$

•**Note that the factor** $\frac{5}{18} = \frac{1}{2}\left(q_u^2 + q_d^2\right)$ **and by comparing neutrino to electron scattering structure functions measure the sum of quark charges**

Experiment: 0.29 ± 0.02

Measurements of $F_2(x)$ and $F_3(x)$

•**CDHS Experiment** $\nu_\mu + \mathrm{Fe} \rightarrow \mu^- + X$



H. Abramowicz et al., Z.Phys. C17 (1983) 283

$$F_2^{\nu N} = x[u(x) + d(x) + \bar{u}(x) + \bar{d}(x)]$$
$$xF_3^{\nu N} = x[u(x) + d(x) - \bar{u}(x) - \bar{d}(x)]$$

$$\Rightarrow F_2^{\nu N} - xF_3^{\nu N} = 2x[\bar{u} + \bar{d}]$$

✶ **Difference in neutrino structure functions measures anti-quark (sea) parton distribution functions**

Sea dominates so expect xF_3 to go to zero as $q(x) = \bar{q}(x)$

Sea contribution goes to zero

Valence Contribution

★ **Separate parton density functions into sea and valence components**

$$u(x) = u_V(x) + u_S(x) = u_V(x) + S(x)$$
$$d(x) = d_V(x) + d_S(x) = d_V(x) + S(x)$$
$$\bar{u}(x) = \bar{u}_S(x) = S(x)$$
$$\bar{d}(x) = \bar{d}_S(x) = S(x)$$

➡ $$F_3^{\nu N} = [u(x) + d(x) - \bar{u}(x) - \bar{d}(x)] = u_V(x) + d_V(x)$$

➡ $$\int_0^1 F_3^{\nu N}(x)\mathrm{d}x = \int_0^1 (u_V(x) + d_V(x))\,\mathrm{d}x = N_u^V + N_d^V$$

★ **Area under measured function $F_3^{\nu N}(x)$ gives a measurement of the total number of valence quarks in a nucleon !**

expect $$\int_0^1 F_3^{\nu N}(x)\mathrm{d}x = 3$$

"Gross – Llewellyn-Smith sum rule"

Experiment: 3.0 ± 0.2

•**Note:** $F_2^{\bar{\nu}p} = F_2^{\nu n};\ F_2^{\bar{\nu}n} = F_2^{\nu p};\ F_3^{\bar{\nu}p} = F_3^{\nu n};\ F_3^{\bar{\nu}n} = F_3^{\nu p}$ **and anti-neutrino structure functions contain same pdf information**

REVIEW Some other Cool Concepts

- Spin Operator (in Griff's)

$$\mathbf{S} = \frac{\hbar}{2}\boldsymbol{\Sigma}, \quad \text{with } \boldsymbol{\Sigma} \equiv \begin{pmatrix} \boldsymbol{\sigma} & 0 \\ 0 & \boldsymbol{\sigma} \end{pmatrix}$$

- Helicity Operator – we can make more explicit

  both of above, other solution sets to Dirac generally

$$\boldsymbol{\Sigma} \cdot \hat{\mathbf{p}} \equiv \begin{pmatrix} \boldsymbol{\sigma} \cdot \hat{\mathbf{p}} & 0 \\ 0 & \boldsymbol{\sigma} \cdot \hat{\mathbf{p}} \end{pmatrix}$$

- **Cool concept**: when E>> m Lorentz boosting must cause states to get into Helicity eigenstates
  - (Sz → large pz) because all directions get boosted into "z direction"
- Also ignoring m Dirac equation simplifies to 2 - 2x2 "helicity"-like eigenstates

5.7 Zero-Mass Fermions: The Two-Component Neutrino

We return to the Dirac equation, (5.1),

$$H\psi = (\boldsymbol{\alpha}\cdot\mathbf{p} + \beta m)\psi$$

We can write the solutions as

$$\psi_L = e^{-ip\cdot x}\, \sigma\cdot p\, \chi$$

$$\bar{\sigma}\cdot\partial = -i\,\bar{\sigma}\cdot p\,\sigma\cdot p\,\chi\, e^{-ip\cdot x} = 0$$

and $\psi_R = e^{-ip\cdot x}\,\bar{\sigma}\cdot p\,\chi$.

$$E\chi = -\boldsymbol{\sigma}\cdot\mathbf{p}\,\chi, \quad \text{"}u_R\text{"} \quad \chi\begin{pmatrix}1\\0\end{pmatrix}$$

$$E\phi = +\boldsymbol{\sigma}\cdot\mathbf{p}\,\phi. \quad \text{"}u_L\text{"} \quad \chi\begin{pmatrix}0\\1\end{pmatrix}$$

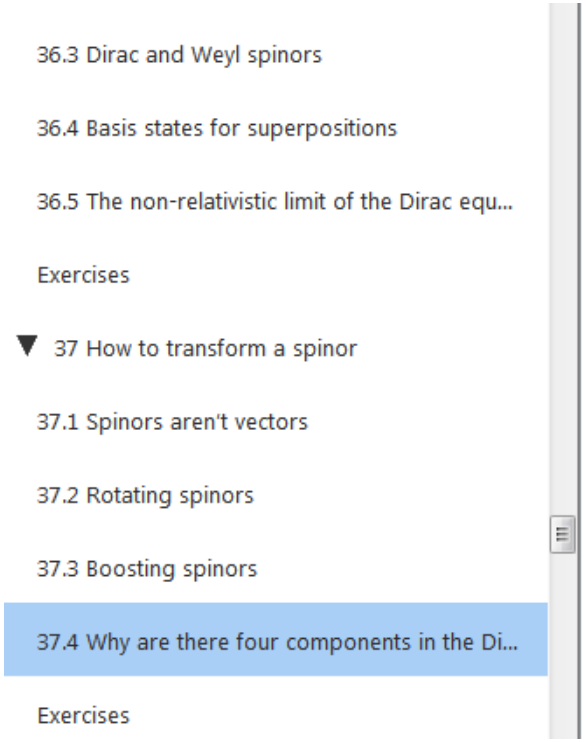
γ^5 → 2x2 Projection operator

$$\tfrac{1}{2}(1-\gamma^5)u_\nu = \begin{pmatrix} I & 0 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} \chi \\ \phi \end{pmatrix} = \begin{pmatrix} \chi \\ 0 \end{pmatrix}$$

Why 4 Complex Components?

- Why did we suddenly insert parity into the conversation:

- See QFT Lancaster/Blundell chapter:
  - In many limits, the 4-D spinor becomes like a repeat of two 2-spinors e.g. $p \to \infty$ (1010) so you may ask :



37.4 Why are there four components in the Dirac equation?

The Dirac equation is written in terms of a four-component Dirac spinor ψ. These Dirac spinors have two degrees of freedom: the two components of ξ (or η). So why do we have four components in the Dirac equation?

We found that for the case of massless particles the Dirac spinors fell apart into two: left-handed Weyl spinors and right-handed Weyl spinors. Each of these spinors carried the same spin information in ξ (or η). The question is, then, why does the world need left- and right-handed fields. We will answer this question by asking what fundamental operation turns left-handed spinors into right-handed spinors? The answer is parity.

Many of the properties of fermions are invariant with respect to parity and so we require that Dirac particles have the same properties in a parity reversed world. (The weak interaction violates parity, but we ignore that for now.) The parity operation P does the following: it maps $x \to -x$ and $v \to -v$ and so it results in the changes to the generators:

$$\mathsf{P}^{-1}\boldsymbol{K}\mathsf{P} = -\boldsymbol{K}, \quad \mathsf{P}^{-1}\boldsymbol{J}\mathsf{P} = +\boldsymbol{J} \tag{37.20}$$

"the answer is parity ..."

Another Interesting Insight

- Reducing the Shrodinger or Klein Gordon Equation to linear in all derivatives results in less constraints
- More degrees of freedom in solution

Common other way to derive Dirac equation/gamma's → start by defining α, β which become $\gamma^{0\,-1-2-3}$ matrix's

The Dirac Equation :

- Schrödinger eqn: $-\frac{1}{2m}\vec{\nabla}^2\psi = i\frac{\partial\psi}{\partial t}$ — 1st order in $\partial/\partial t$; 2nd order in $\partial/\partial x, \partial/\partial y, \partial/\partial z$
- Klein-Gordon eqn: $(\partial^\mu\partial_\mu + m^2)\psi = 0$ — 2nd order throughout
- Dirac looked for an alternative which was 1st order throughout:

$$\hat{H}\psi = (\vec{\alpha}.\vec{p} + \beta m)\psi = i\frac{\partial\psi}{\partial t} \qquad \text{(D1)}$$

where $\hat{H}$ is

- Writing (D1) i

"squaring" t

$$\left(-i\alpha_x\frac{\partial}{\partial x} - i\alpha_y\right.$$

- Which can be

Prof. M.A. Thomson

- Consequently the wave-function must be a four-component Dirac Spinor

$$\psi = \begin{pmatrix}\psi_1\\ \psi_2\\ \psi_3\\ \psi_4\end{pmatrix}$$

A consequence of introducing an equation that is 1st order in time/space derivatives is that the wave-function has new degrees of freedom !

- For the Hamiltonian $\hat{H}\psi = (\vec{\alpha}.\vec{p} + \beta m)\psi = i\partial\psi/\partial t$ to be Hermitian requires

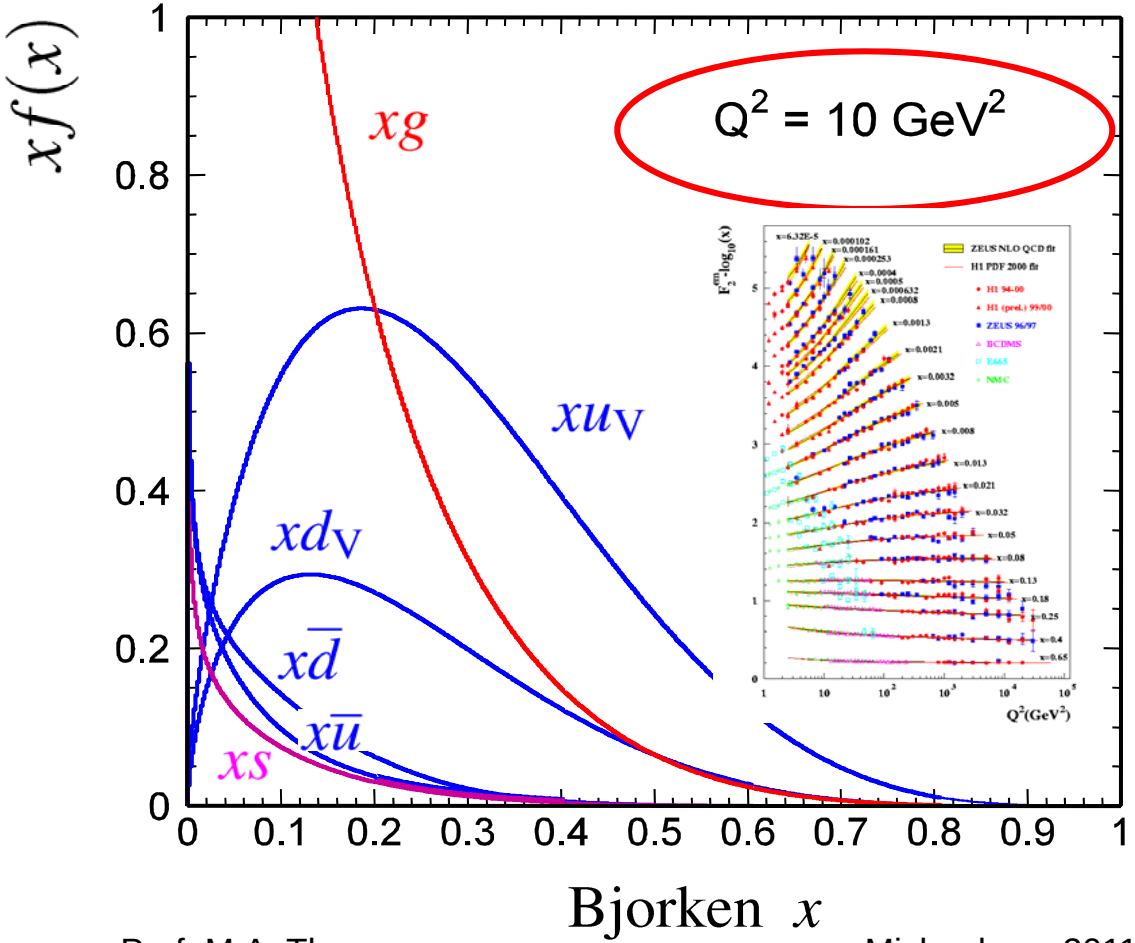
$$\alpha_x = \alpha_x^\dagger; \quad \alpha_y = \alpha_y^\dagger; \quad \alpha_z = \alpha_z^\dagger; \quad \beta = \beta^\dagger; \qquad \text{(D5)}$$

i.e. the require four anti-commuting Hermitian 4x4 matrices.

- At this point it is convenient to introduce an explicit representation for $\vec{\alpha}$, β

Universality of PDF's→ Need for Evolution

- MAIN POINT : The info we get on the pdf(x)'s is expected to be UNIVERSAL ! Once measured in one situation they can apply to any other situation.

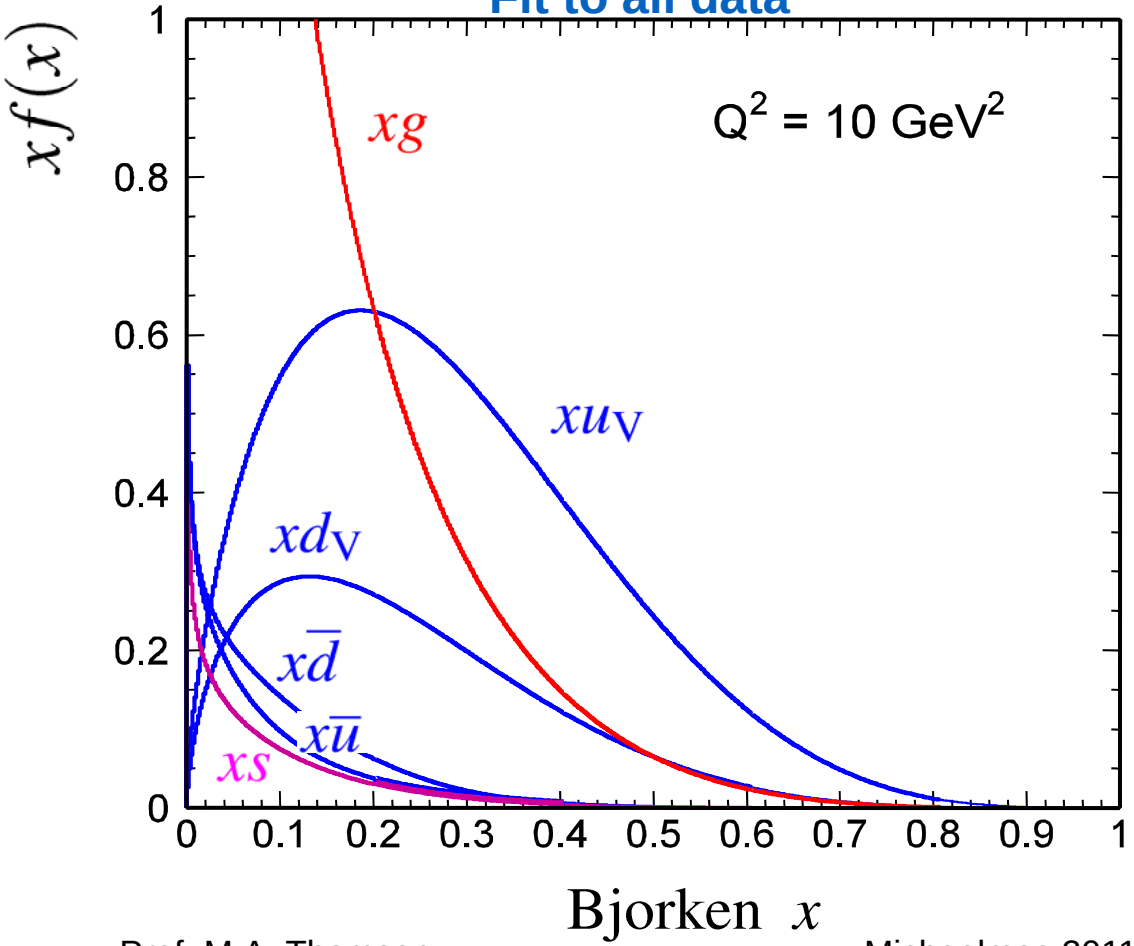


- Generally measure in **discrete** bins of Q^2 and x in DIS
- But even to use these in more complicated situations even in e+p
  - Need continuous Q,x dependence!
- Hence need for EVOLUTION

Parton Distribution Functions

- **Ultimately the parton distribution functions are obtained from a fit to all experimental data including neutrino scattering (see handout 10)**
  - **Hadron-hadron collisions give information on gluon pdf** $g(x)$

Fit to all data



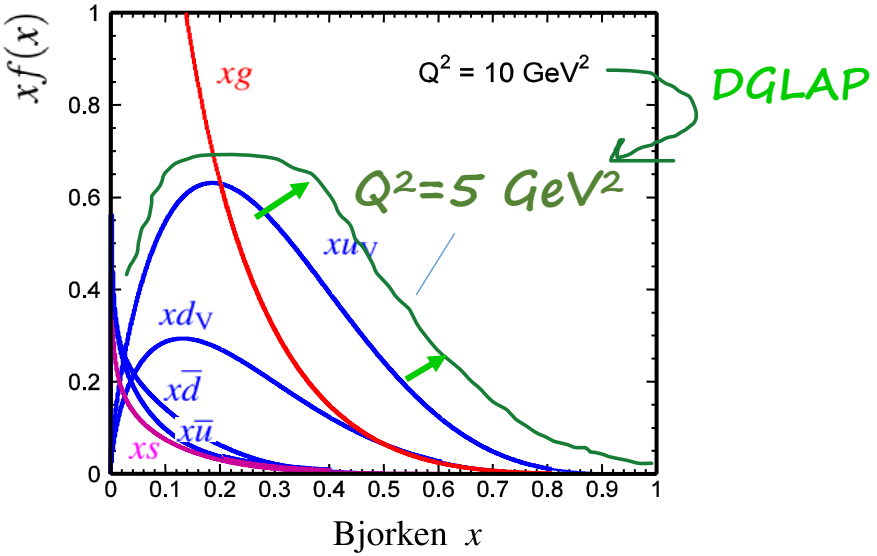
Note:

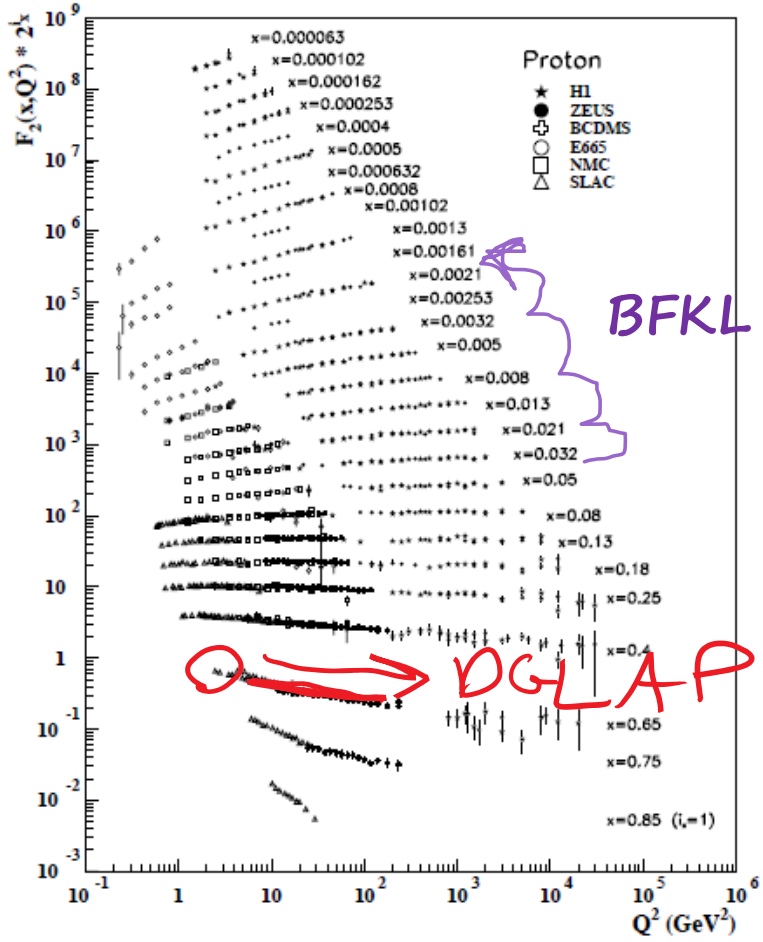
- **Apart from at large** x
  $$u_V(x) \approx 2d_V(x)$$
- **For** $x < 0.2$
  gluons dominate
- **In fits to data assume**
  $$u_s(x) = \bar{u}(x)$$
- $\bar{d}(x) > \bar{u}(x)$
  not understood – exclusion principle?
- **Small strange quark component** $s(x)$

(Try Question 12)

DGLAP Evolution / BFKL

- So DGLAP Evolution theory can take MEASURED VALUE of either F structure fn's or pdf's themselves especially gluon pdf g(x) at any Q1^2 and accurately predict the same fn's for another Q^2 value (e.g. Q2^2)
  - "BFKL" is similar type of evolution but in x, i.e. at fixed Q^2, takes F(x1) to F(x2)
- So we can take perturbative calcs in regime of hadron structure but need "starter" data





DGLAP Struct Fn (& pdf) Evolution

- DGLAP pQCD "Evolution" takes f(x,q1^2) to another q2
- Uses q/g "splitting functions" Can be predicted from **Perturbative** QCD

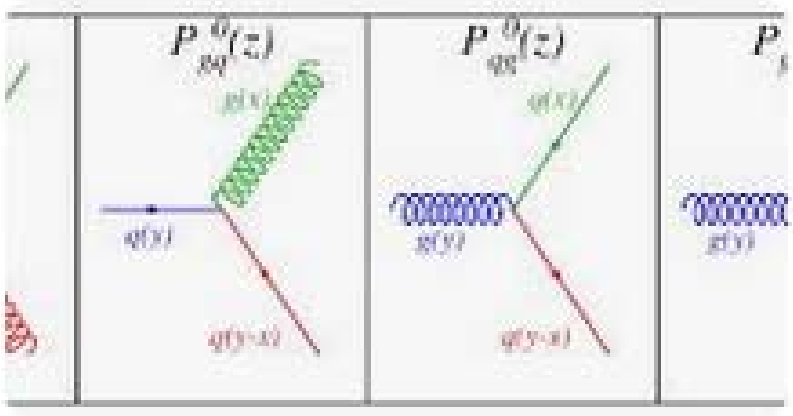
$$\frac{dq(x,Q^2)}{d\ln Q^2} = \frac{\alpha_s(Q^2)}{2\pi}\int_0^1 \frac{dy}{y}\left[\Sigma_q P_{qq}(z)q(y,Q^2) + P_{qg}(z)g(y,Q^2)\right]$$

$$P_{qg} \to \frac{C_F}{z}, \quad P_{gg} \to \frac{2C_A}{z}$$

$$\frac{dq(x,Q^2)}{d\ln Q^2} = \frac{\alpha_s(Q^2)}{2\pi}\int_0^1 \frac{dy}{y}\frac{6}{z}g(y,Q^2)$$

Alt.Parisi(AP) Evolution diffeq (DGLAP solves)

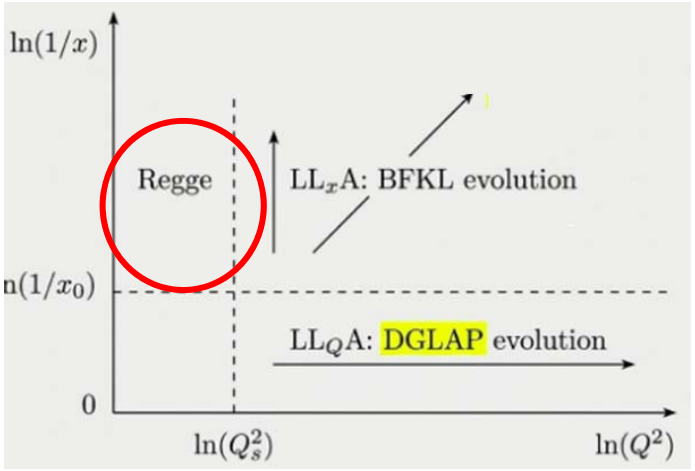
$$\begin{aligned}
P_{q_iq_j}(z,\alpha_s) &= \delta_{ij}P_{qq}^{(0)}(z) + \frac{\alpha_s}{2\pi}P_{q_iq_j}^{(1)}(z) + \ldots \\
P_{qq}(z,\alpha_s) &= P_{qq}^{(0)}(z) + \frac{\alpha_s}{2\pi}P_{qq}^{(1)}(z) + \ldots \\
P_{qg}(z,\alpha_s) &= P_{qg}^{(0)}(z) + \frac{\alpha_s}{2\pi}P_{qg}^{(1)}(z) + \ldots \\
P_{gg}(z,\alpha_s) &= P_{gg}^{(0)}(z) + \frac{\alpha_s}{2\pi}P_{gg}^{(1)}(z) + \ldots
\end{aligned}$$

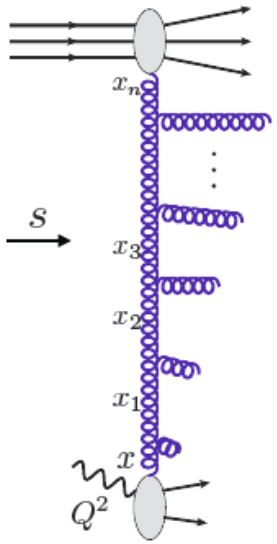


- BFKL evolution is similar thing but x1→x2 evolution

BFKL equations

- BFKL x evolution uses pqcd treatment of increasing numbers of partons → (2→ 3, etc)
- Tied to experimental data on rapidity distributions of soft particle production
- All of this physics is important to low Q^2 diffraction and Pomeron→ next wk





Resummation performed by BFKL evolution equation

$$\frac{\partial \mathcal{F}_g(x,k_T)}{\partial \ln 1/x} = \int d^2k'_T\, \mathcal{K}(k_T,k'_T)\, \mathcal{F}_g(x,k'_T)$$

Branching kernel (perturbative expansion)

$$\mathcal{K} = \bar{\alpha}_s \mathcal{K}^{LLx} + \bar{\alpha}_s^2 \mathcal{K}^{N}$$

QCD

$$\frac{dN(x_{01},b_t,y)}{dy} = -\frac{2C_F\alpha_S}{\pi}\ln\left(\frac{x_{01}^2}{\rho^2}\right) N(x,b_t,y) + \frac{2C_F\alpha_S}{\pi}\int_\rho d^2x_2 \frac{x_{01}^2}{x_{02}^2 x_{12}^2}\cdot N(x_{02},b_t,y)$$

where

$$\mathcal{F}_g(x,k_T)$$

Unintegrated, (transverse momentum dependent) gluon density

plots/slides from A. Statso and Bondarenko

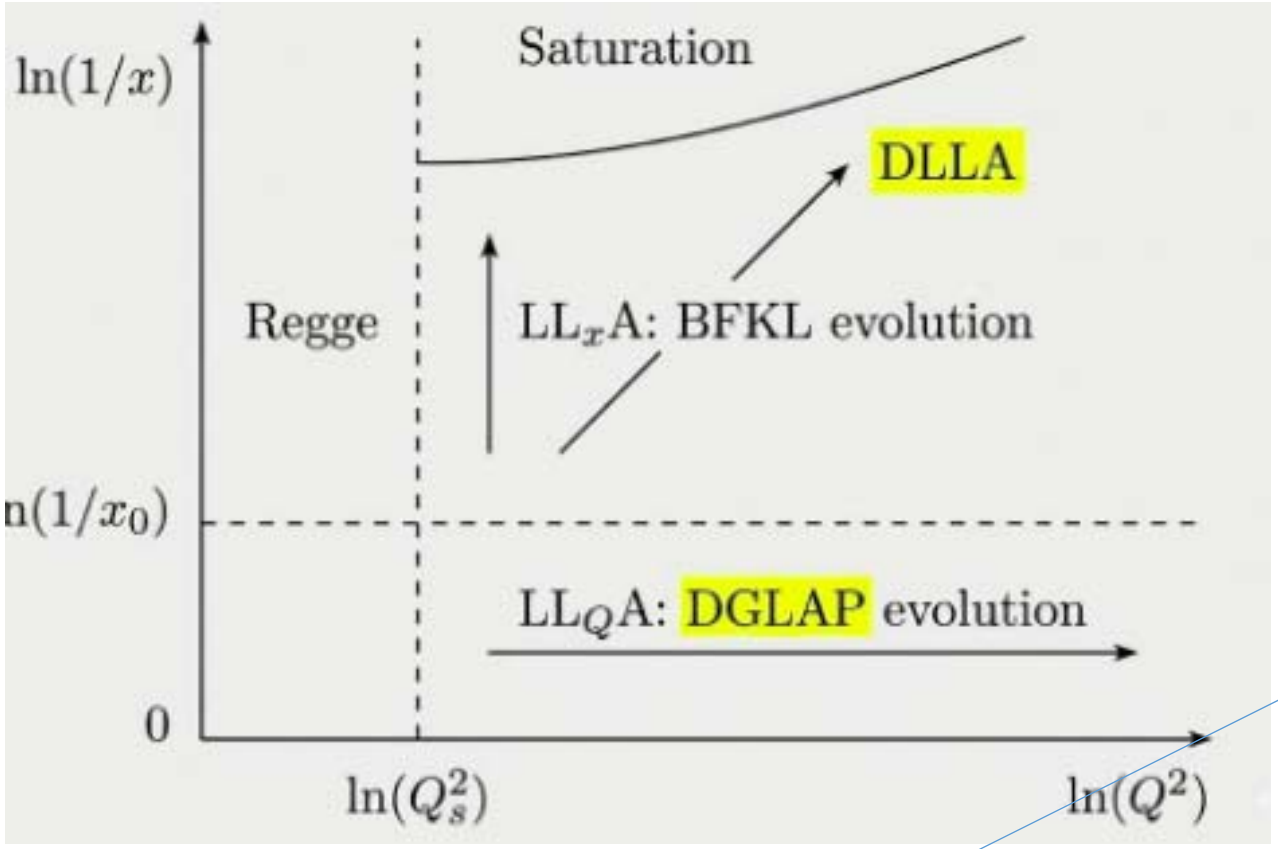
This is our y (rapidity!)

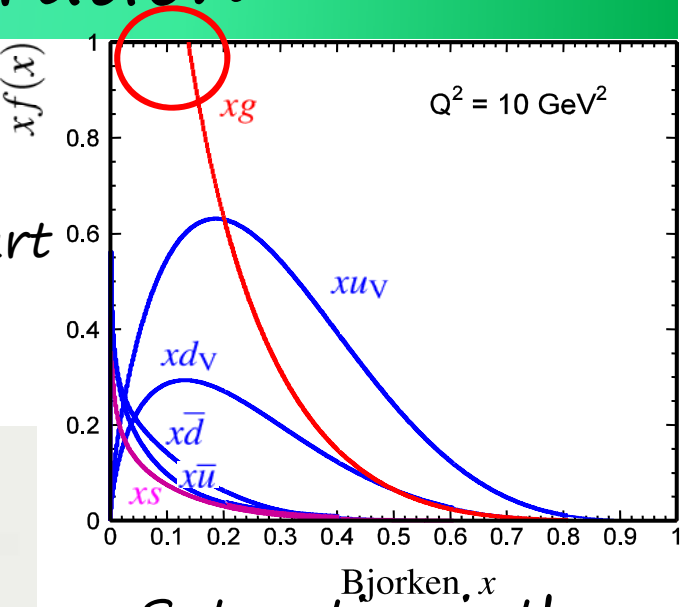
compare with DGLAP-collinear approach

$$\frac{\partial f_i(x,Q^2)}{\partial \log(Q^2)} = \sum_j \int_x^1 \frac{dz}{z} P_{j\to i}(z) f_j\left(\frac{x}{z},Q^2\right)$$

DGLAP/BFKL→ Saturation

- "Saturation" in small x, lower Q^2
- g(x) gets too BIG!!! "breaks unitarity"
  - But we know low E gluons would eventually start interacting→ naturally quenches g(x) @ x→0



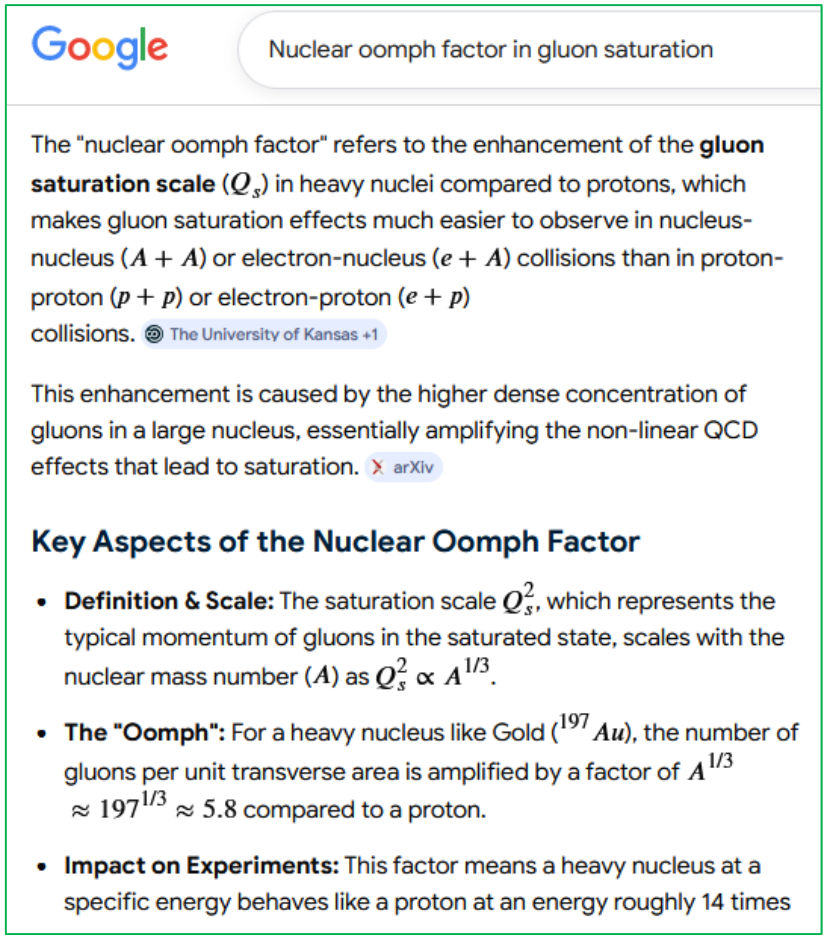


- Saturation is the general concept that that would happen
- Various models based on
  - BK-BFKL
- One of the most developed → "Color Glass Condensate" an effective Field Theory

Yuri Kovechgov Ohio State nearby!

Saturation in Nuclei—OOMPH!!!

- NUCLEI (as opposed to single p,n) → expected to have their own g(x) not associated with p's or n's
- Another reason Saturation is a motivation for EIC @ BNL → electron ION



Google | Nuclear oomph factor in gluon saturation

The "nuclear oomph factor" refers to the enhancement of the **gluon saturation scale** (Q_s) in heavy nuclei compared to protons, which makes gluon saturation effects much easier to observe in nucleus-nucleus ($A + A$) or electron-nucleus ($e + A$) collisions than in proton-proton ($p + p$) or electron-proton ($e + p$) collisions. The University of Kansas +1

This enhancement is caused by the higher dense concentration of gluons in a large nucleus, essentially amplifying the non-linear QCD effects that lead to saturation. arXiv

Key Aspects of the Nuclear Oomph Factor

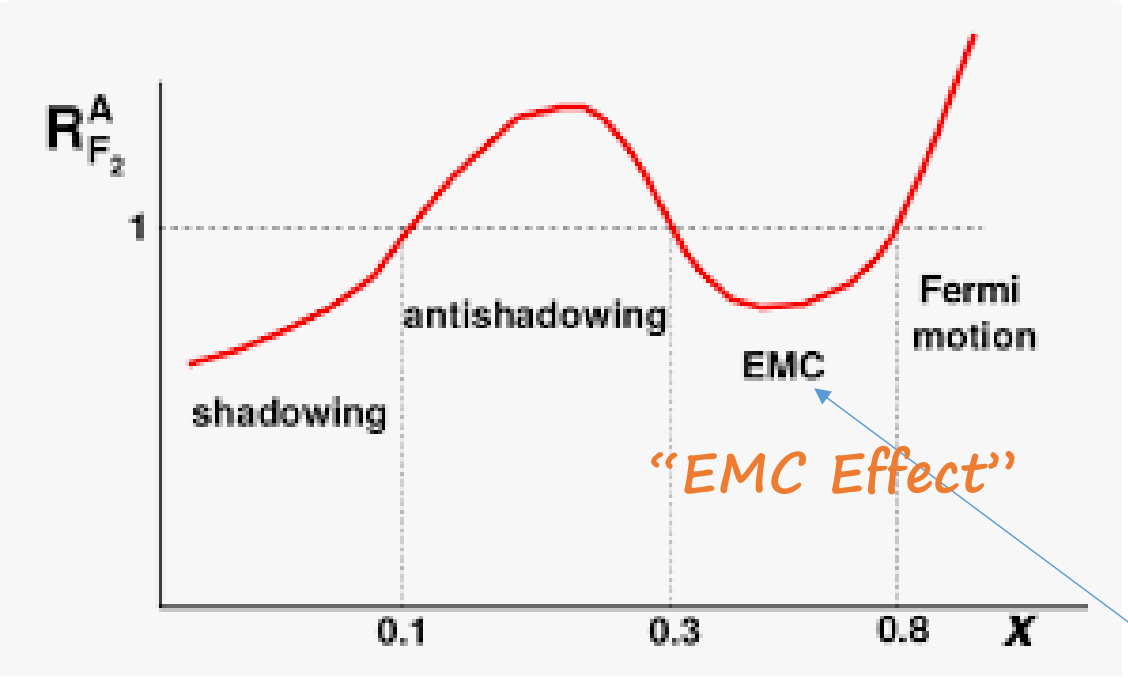
- **Definition & Scale:** The saturation scale Q_s^2, which represents the typical momentum of gluons in the saturated state, scales with the nuclear mass number (A) as $Q_s^2 \propto A^{1/3}$.
- **The "Oomph":** For a heavy nucleus like Gold (^{197}Au), the number of gluons per unit transverse area is amplified by a factor of $A^{1/3} \approx 197^{1/3} \approx 5.8$ compared to a proton.
- **Impact on Experiments:** This factor means a heavy nucleus at a specific energy behaves like a proton at an energy roughly 14 times

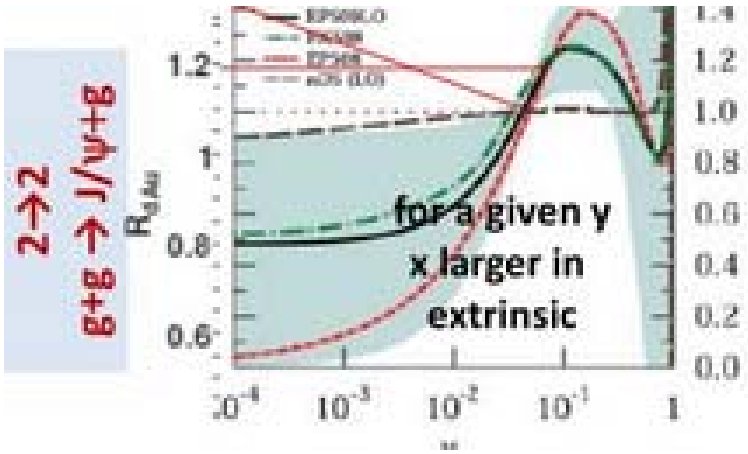
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- Saturation onsets A^1/3 larger Q^2 (QS – "saturation scale")

Nuclear PDF's

- Generally we expect PDF's to be a little different when p,n embedded in nuclei (At all x -- saturation makes it VERY different for low x g(x))
- nPDF's. Similar global fits
- You would expect it would also be something for EIC !!!
  - HERA DIDN''T HAVE THAT!!!!!



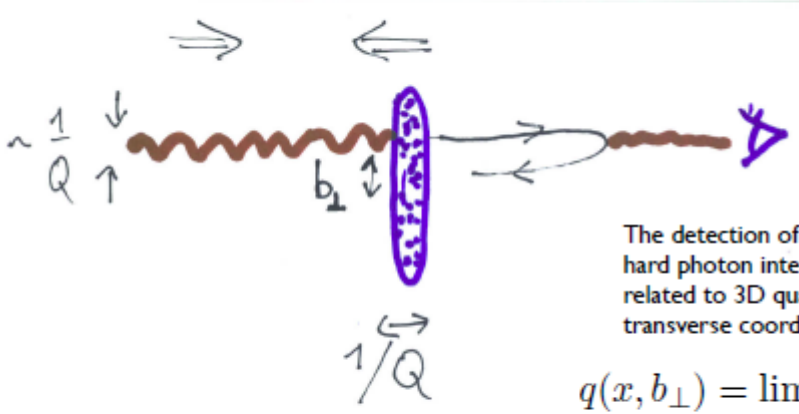


There is evidence from measurements @ JLAB it's from "Short Range Correlations" SRC relevant for current Neutron Star calcs

GPDs Generalized Parton Distributions

- Next gen pdf's : 2D/3D rather than collinear

Relation of GPDs to 3D image of hadrons



The detection of the final photon allows to find 3D location of the hard photon interaction with an (anti) quark. Therefore GPDs are related to 3D quark density (in longitudinal momentum and in transverse coordinate spaces)

$$q(x, b_\perp) = \lim_{\xi \to 0} \int \frac{d^2\Delta_\perp}{(2\pi)^2} \, e^{i\Delta_\perp \cdot b_\perp} \, H(x, \xi, -\Delta_\perp^2)$$

Nice, but there are problems!

- This limit is not achievable in an experiment
- The variable x is integrated out in the amplitude.

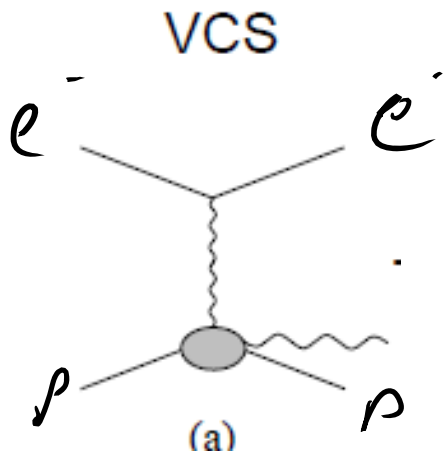
$H(x, \xi, t)$

these are external kinematical variables

$$A(\xi, t) = \int_{-1}^{1} dx \, \frac{H(x, \xi, t)}{x - \xi + i0}$$

DVCS - GPD's

- Extreme precision e+p



$$H^q(x,0,0) = \begin{cases} q(x), & x > 0, \\ -\bar{q}(-x), & x < 0. \end{cases}$$

$$\tilde{H}^q(x,0,0) = \begin{cases} \Delta q(x), & x > 0, \\ \Delta\bar{q}(-x), & x < 0. \end{cases}$$

H(x) and E(x) are generalizations of pdfs which encode all of their info, plus a lot more (e.g. correlations)

Generalized Parton Dist

$\gamma^* p \to \gamma p$

pseudo-elastic similar to pomeron

$$\int_{-1}^{+1} dx\, H^q(x,\xi,t) = F_1^q(t),$$

$$\int_{-1}^{+1} dx\, E^q(x,\xi,t) = F_2^q(t),$$

$$\int_{-1}^{+1} dx\, \tilde{H}^q(x,\xi,t) = g_A^q(t),$$

$$\int_{-1}^{+1} dx\, \tilde{E}^q(x,\xi,t) = g_P^q(t).$$

Factorization- Universality

$$\sigma \sim \iiint F(x_1) \;\; F(x_2) \;\; \hat{\sigma} \;\; D(z_1) \;\; D(z_2).$$

- Factorization: the above integral– proven mathematically. Different versions for different processes above is for hadron-hadron goes to 2-jet.
  - pdfs (the F() 's) : for both incoming 2 partons
  - **New: Fragmentation Functions:** the D()'s: The REVERSE kind of function called Fragmentation Functions that describe probability distribution of different hadrons carrying fraction z of struck parton (jet) momentum
- Idea for PDF's,FF's is should be "universal"
  - ~Independent of Q^2 (turns out not to be true, but QCD can predict why exactly)
  - **Independent of process** (this too has run into problems as well, .e.g e+p vs. mu+p, vs. nu+p, but has been fixable)

- In some historical situations, without higher orders, pQCD's predictive power is limited but lately these uncertainties are better understood → Higg's discovery required precision knowledge of pdfs